



CHAPTER 06

Conditional Expression



- Sometimes we want to play **PUBG** on our phone if the day is **Sunday**.
- Sometimes we order **Ice Cream** online if the day is **sunny**.
- Sometimes we go **hiking** if our parents **allow**.
- All these are **decisions** which depend on a **condition** being met.
- In python programming too, we must be able to execute instructions on a **condition(s)** being met.
- This is what **conditionals** are for!



Decisions make our program smart!

★ If Else And Elif In Python

If else and elif statements are a **multiway decision** taken by our program due to certain conditions in our code.

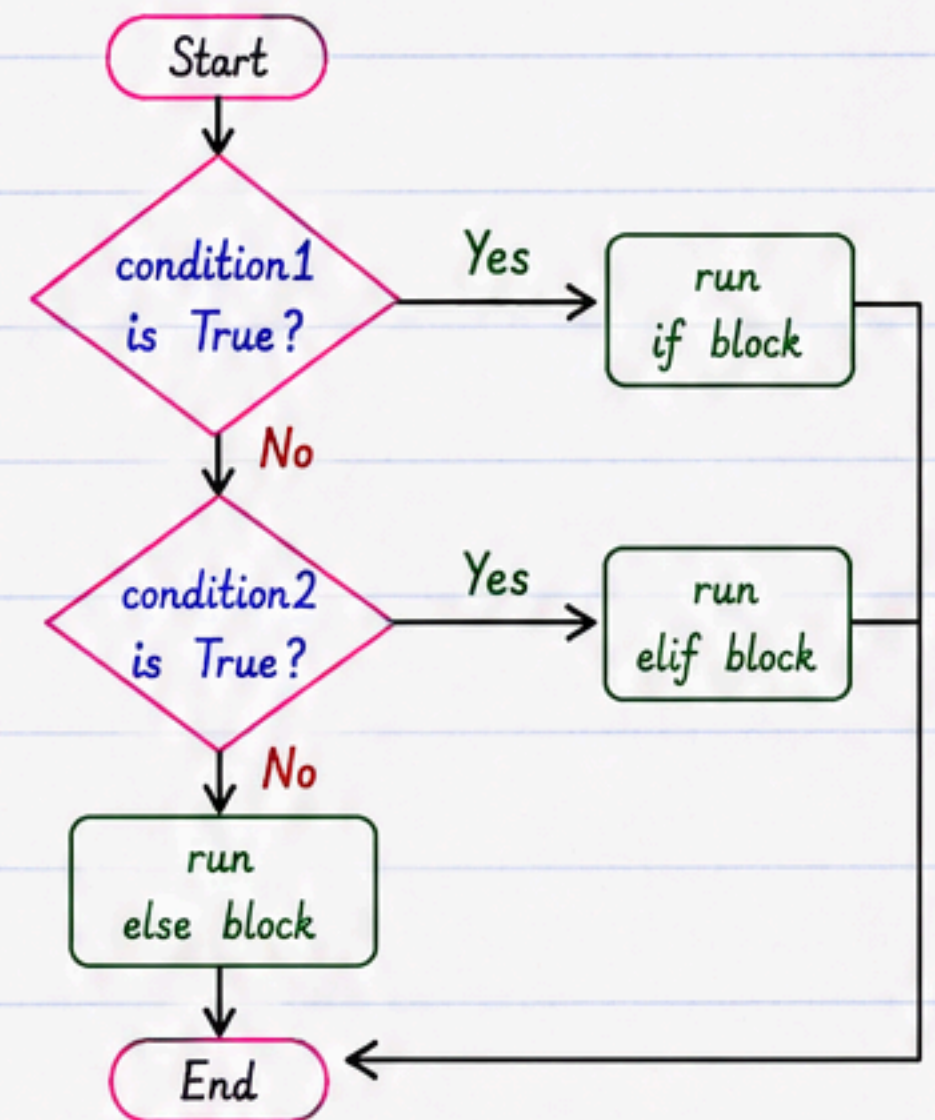
Syntax:

```
if(condition1): # if condition1 is True
    print("yes")
elif(condition2): # if condition2 is True
    print("no")
else:            # otherwise
    print("maybe")
```

Code Example:

```
a = 22
if(a > 9):
    print("greater")
else:
    print("lesser")
```

a = 22
Is 22 > 9 ? → Yes (True)
↓
Output : greater

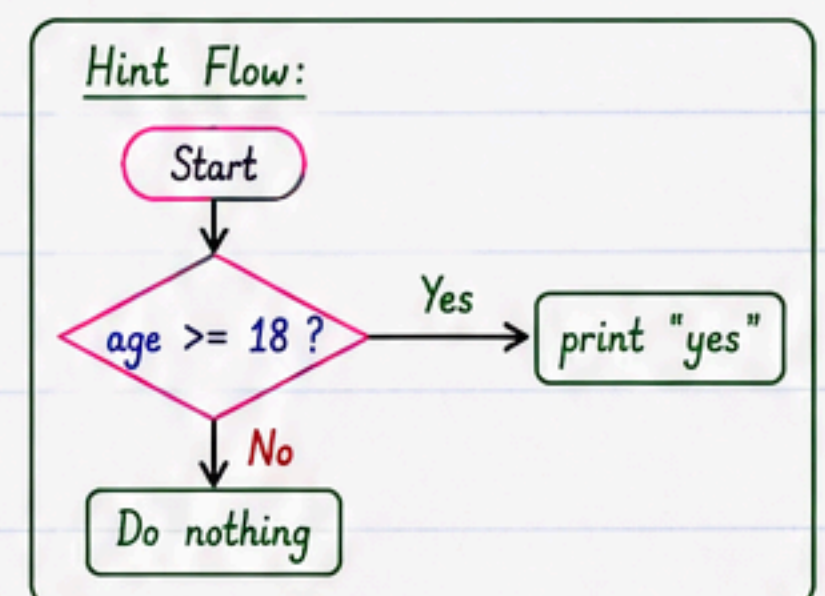


★ Quick Quiz:

Write a program to print **yes** when the age entered by the user is **greater than or equal to 18**.



Conditionals help us make **smart** decisions!





Relational Operators

Relational Operators are used to evaluate conditions inside the if statements. Some examples of relational operators are:

Operator	Meaning
<code>==</code>	equals.
<code>>=</code>	greater than/ equal to.
<code><=</code>	lesser than/ equal to.

Used in conditions



Examples

- `x == y` → True if x is equal to y
- `a >= b` → True if a is greater than or equal to b
- `m <= n` → True if m is lesser than or equal to n



Logical Operators

In python logical operators operate on conditional statements. For Example:

- `and` - true if both operands are true else false.
- `or` - true if at least one operand is true or else false.
- `not` - inverts true to false & false to true.

p	q	p and q	p or q	not p
T	T	T	T	F
T	F	F	T	F
F	T	F	T	T
F	F	F	F	T

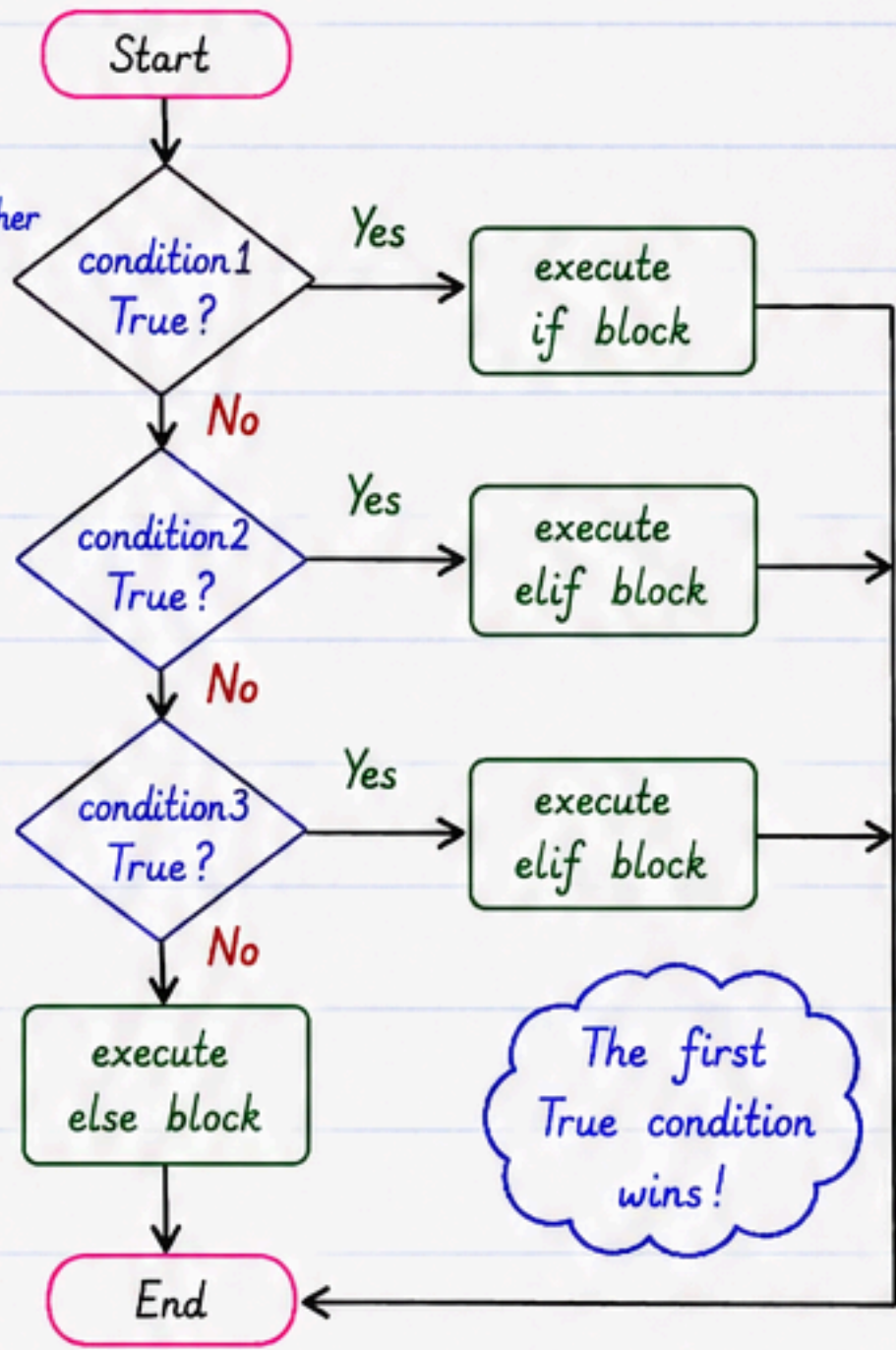
Think before you code!



Elif Clause

elif in python means [else if]. An if statements can be chained together with a lot of these elif statements followed by an else statement.

```
if (condition1):
    #code
elif (condition2): # this ladder will stop
    #code           once a condition in an if or elif is met.
elif (condition3):
    #code
else:
    #code
```



The first True condition wins!

Important Notes:

- There can be any number of elif statements.
- Last else is executed only if all the conditions inside elifs fail.

Practice more, code better!





Practice Set

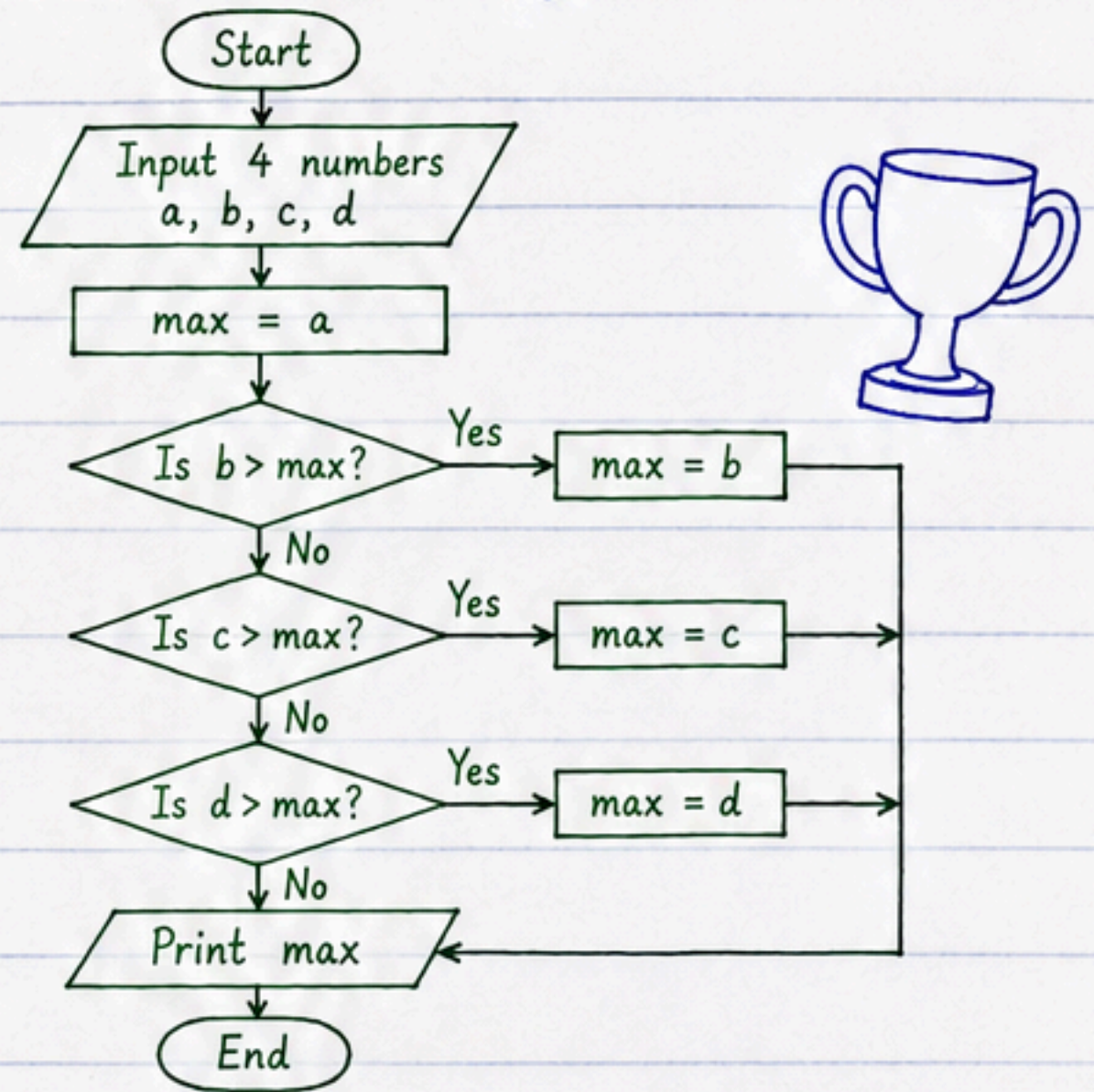


1. Write a program to find the greatest of four numbers entered by the user.

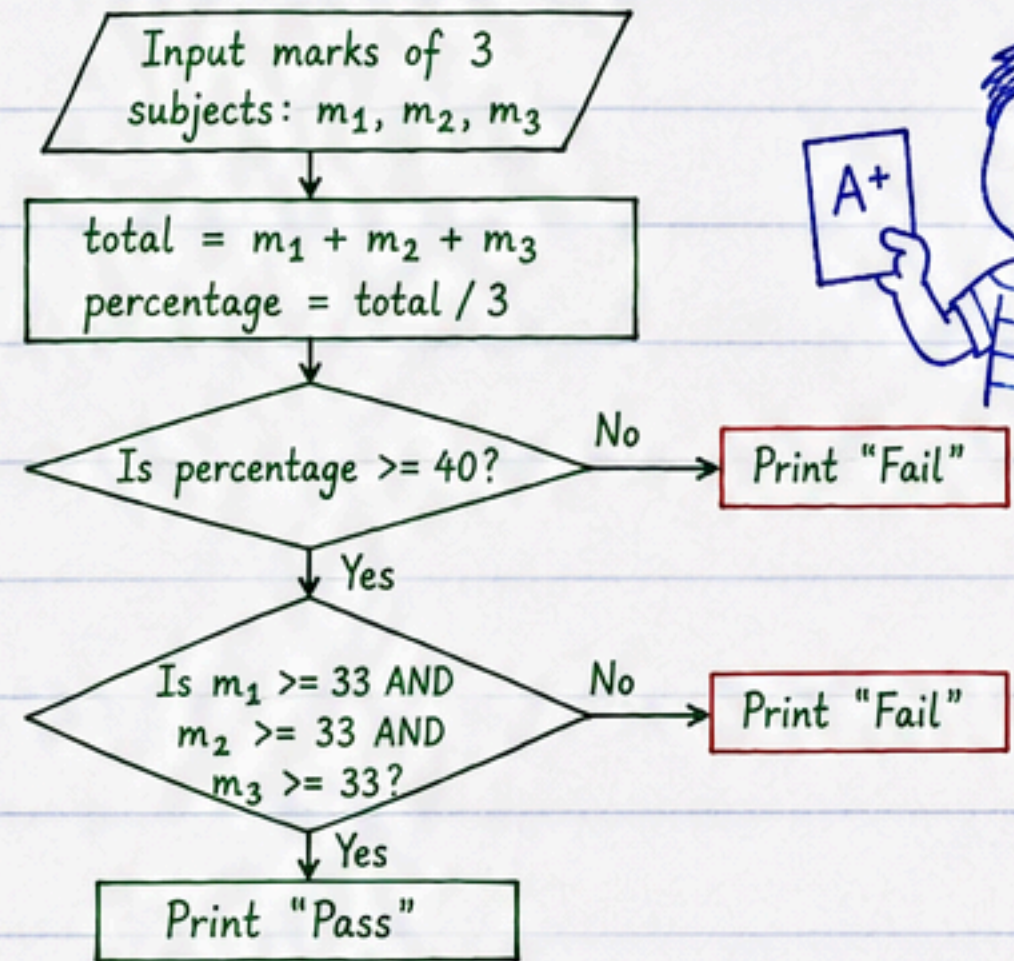
Example:

Input: 12, 45, 7, 29

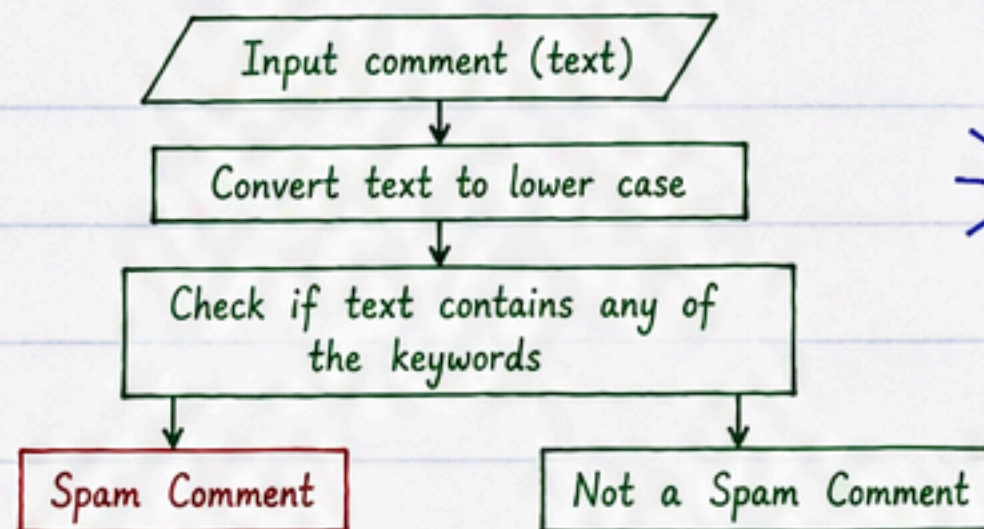
Output: 45



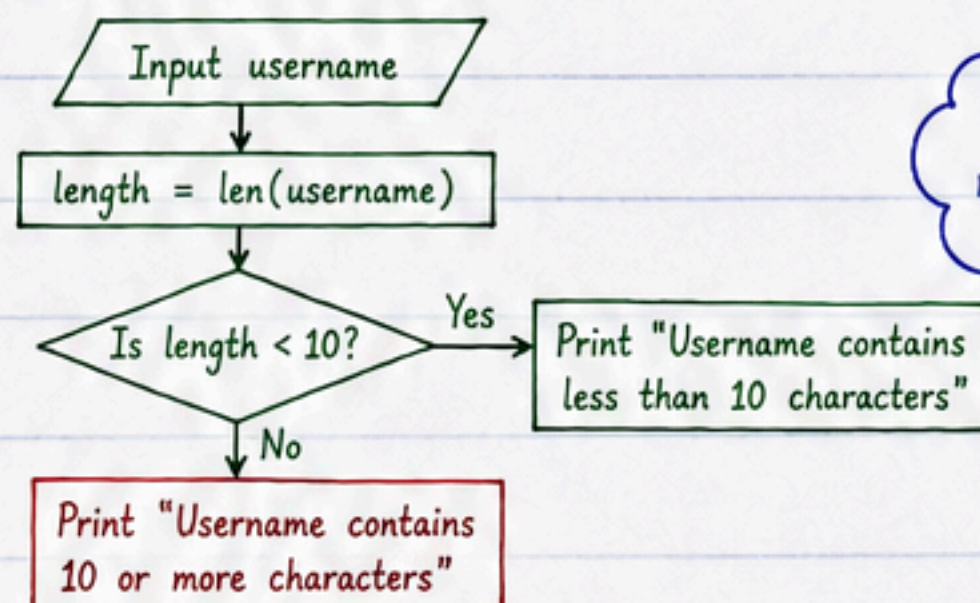
2. Write a program to find out whether a student has passed or failed if it requires a total of 40% and at least 33% in each subject to pass. Assume 3 subjects and take marks as an input from the user.



3. A spam comment is defined as a text containing following keywords: "Make a lot of money", "buy now", "subscribe this", "click this". Write a program to detect these spams.



4. Write a program to find whether a given username contains less than 10 characters or not.



len() function returns the length of the string.





Practice Set (Continued)



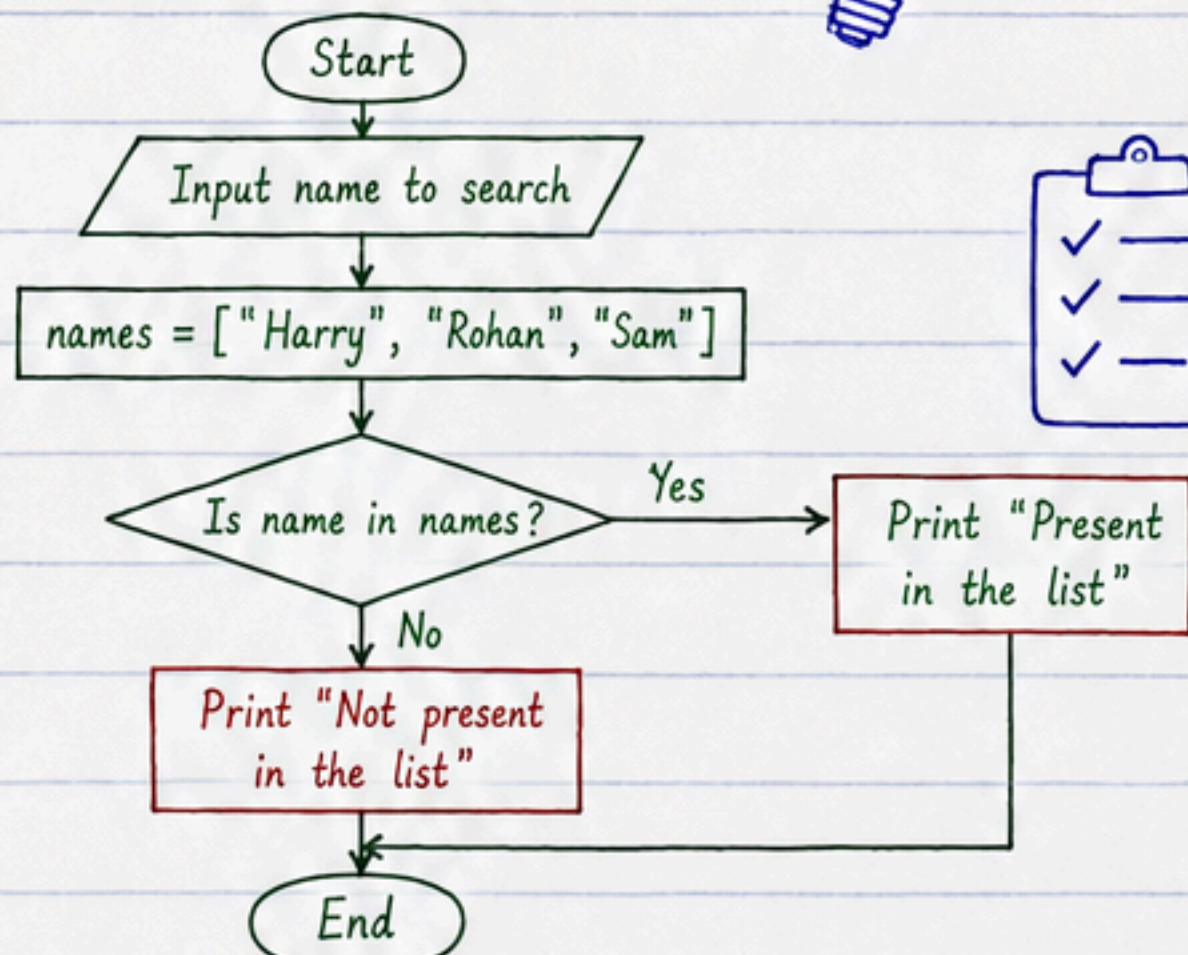
5. Write a program which finds out whether a given name is present in a list or not.

Example:

List: ["Harry", "Rohan", "Sam"]

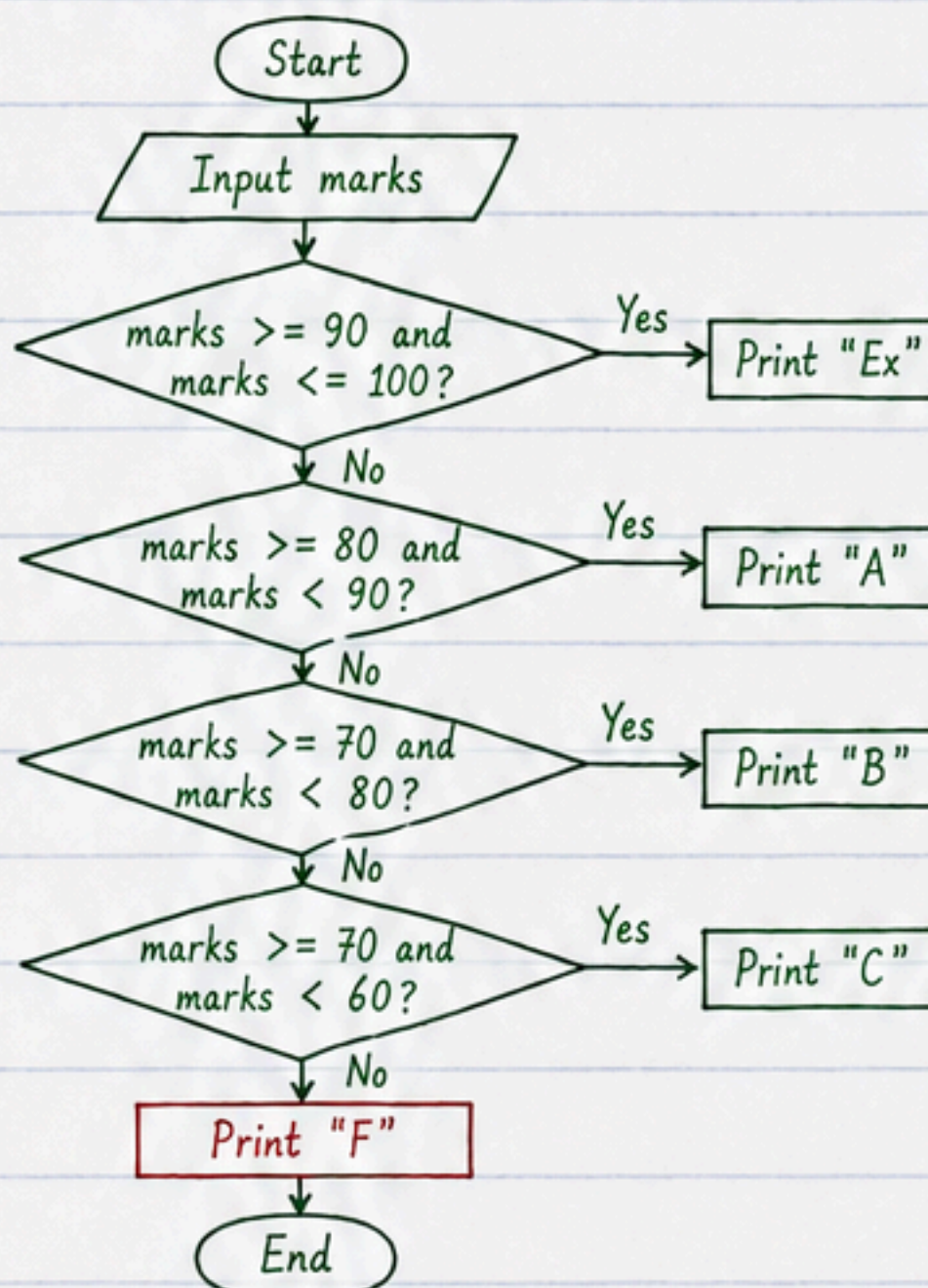
Name to search: "Harry"

Output: Present in the list



6. Write a program to calculate the grade of a student from his marks from the following scheme:

Marks Range	Grade
90 - 100 ⇒	Ex
80 - 90 ⇒	A
70 - 80 ⇒	B
60 - 70 ⇒	C
50 - 60 ⇒	D
< 50 ⇒	F



7. Write a program to find out whether a given post is talking about "Harry" or not.

Example:

Post: "I am learning Python with Harry."

Output: The post is talking about Harry.

Example:

Post: "Python is a great language."

Output: The post is not talking about Harry.

